

FLAME-SENSING FIRE ALARM

Using An Arduino Nano

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Fire safety alarms are crucial in both residential and industrial environments. Early fire detection can prevent significant damage and save lives. Flame sensors offer one of the most efficient methods of fire detection. Here we outline how to construct a simple and reliable fire alarm system using a flame

sensor and an Arduino Nano board.

This fire alarm is intended for hobbyists and individuals interested in learning the basics of electronic sensors and alarms. The simplicity of this device can pave the way for creating life-saving systems. Fig. 1 shows the author's prototype on a breadboard.

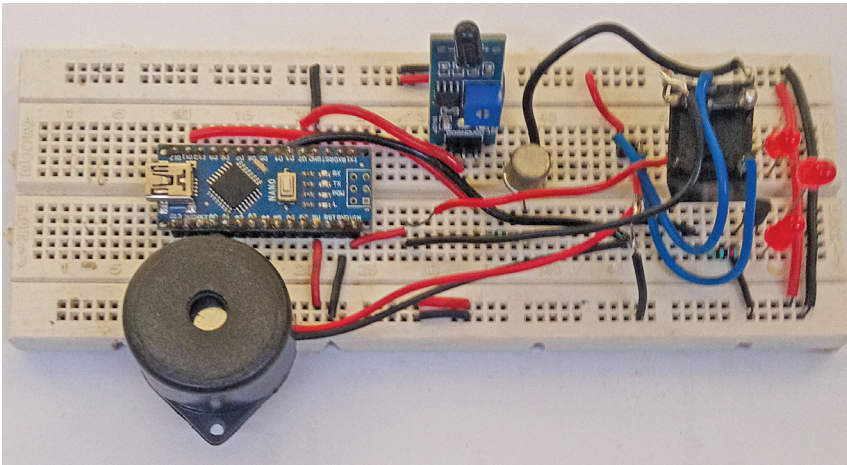


Fig. 1: Author's prototype

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Flamesensor

/* Fire Alarm using Flame Sensor and Arduino Nano
   created by the SURESH DWIVEDI.
   EFY ENTERPRISES PVT LTDHome
*/

int Hooter = 11; // Use buzzer for alert
int FlamePin = 2; // This is for input pin
int Flame = HIGH; // HIGH when FLAME Exposed
int LED = 4; // Use LED for visual alert

void setup() {
  pinMode(Hooter, OUTPUT);
  pinMode(FlamePin, INPUT);
  pinMode(LED, OUTPUT);
  Serial.begin(9600);
}

void loop() {
  Flame = digitalRead(FlamePin);
  if (Flame == LOW)
  {
    Serial.println("HIGH FLAME");
    digitalWrite(LED, HIGH);
    digitalWrite(Hooter, HIGH);
  }
  else
  {
    Serial.println("No flame");
    digitalWrite(LED, LOW);
    digitalWrite(Hooter, LOW);
  }
}

```

Fig. 3: Snippet of the source code

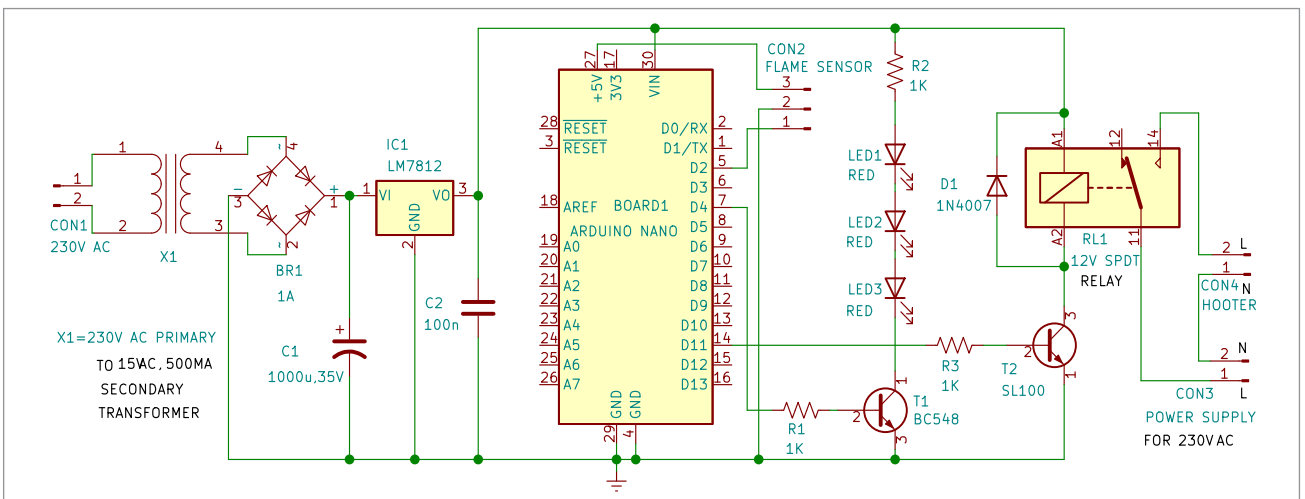


Fig. 2: Circuit diagram